Technical Document #565: Data Engineering - Comprehensive Overview

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Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loa

- Partitioning divides data into smaller, more manageable pieces.

- Data quality ensures data is accurate, complete, and consistent.